Signature detection in documents

Upload a document image. Receive a PNG with signature boxes and confidence scores.

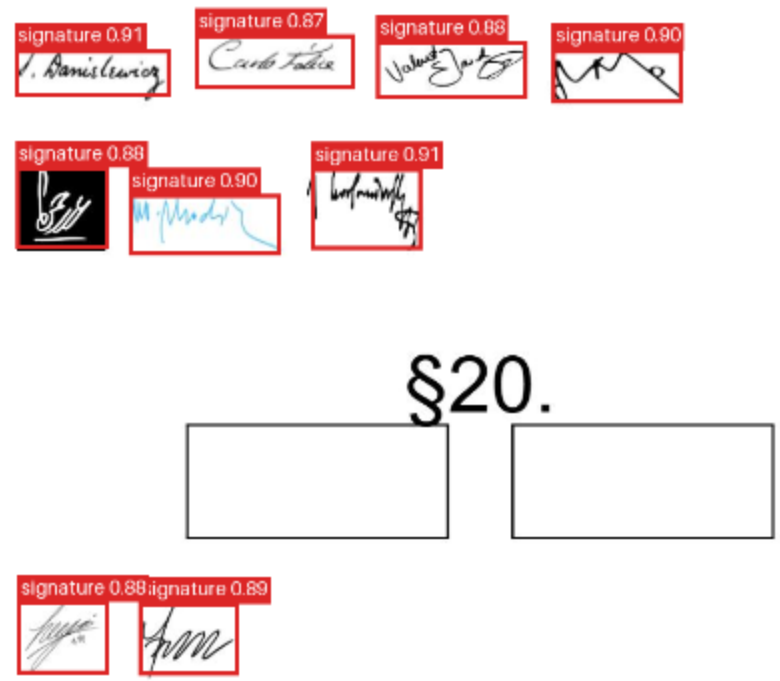
Local data, cleaned and grouped

3 datasets · 3,970 images · 7,859 boxes
RGB conversion, box validation and repair
Related images grouped before splitting

Train 3,176 / Validation 397 / Test 397

Training on the laptop GPU

COCO transfer learning · 416 × 416 input
30 epochs · batch 8 · mixed precision
RTX 3060 Laptop GPU, 6 GB



Held-out example: 9 detections
Signature location; signer identity is not inferred.

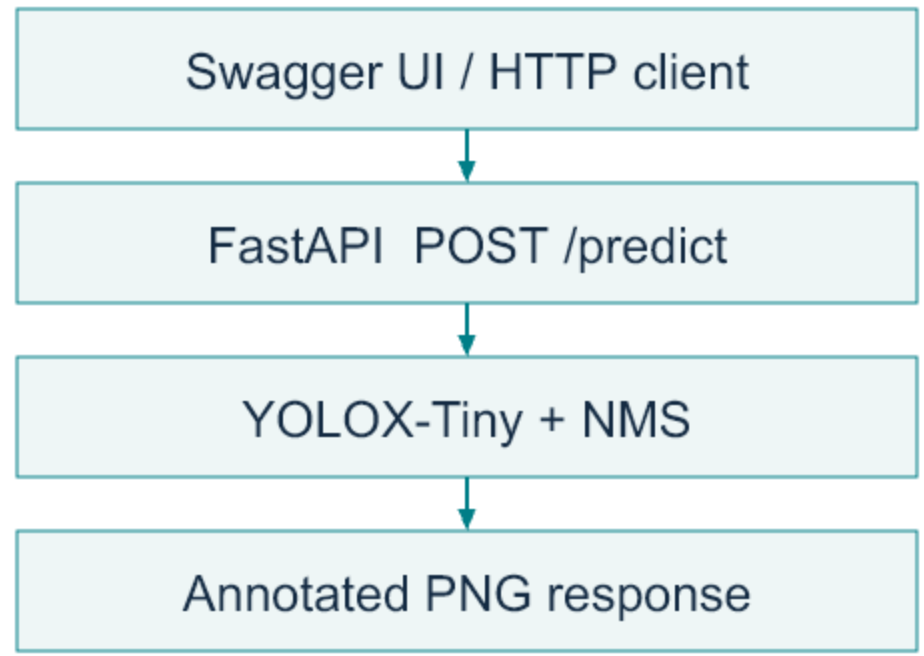
Evaluation and serving architecture

Held-out test: 397 images, 784 signatures

Metric	Result
mAP @ IoU 0.50	**95.03%**
mAP @ IoU 0.50–0.95	**65.51%**
Precision	**92.81%**
Recall	**93.88%**
F1 score	**93.34%**

Precision / recall / F1: confidence 0.25, IoU 0.50.
736 true positives · 57 false positives · 48 missed boxes.

Performance on new document collections requires validation.



Docker · CPU inference
Bundled 20.3 MB model
One worker · two PyTorch CPU threads